Exploring Effects of Shoulder-Assist Exoskeletons on Load-Carrying Stability: A Phase-Specific Approach

Yuanhao Chen[1], Neethan Ratnakumar[2], Mathew Rodriguez[1], Mariya Tohfafarosh[2], Liying Zheng[2], Christopher Pan[3], Xianlian Zhou[2], Wei Yin[1,4]*

[1] Department of Mechanical and Industrial Engineering, New Jersey Institute of Technology, Newark, NJ, USA

[2] Department of Biomedical Engineering, New Jersey Institute of Technology, Newark, NJ, USA

[3] National Institute for Occupational Safety and Health, Morgantown, WV, USA

[4] School of Applied Engineering and Technology, New Jersey Institute of Technology, Newark, NJ, USA

Manual material handling (MMH), a common task in occupational settings that frequently requires workers to transition immediately from lifting to short-distance load-carrying, is associated with a high risk of musculoskeletal injuries. Although commercial exoskeletons have shown potential to reduce local physical workload during isolated lifting tasks, their effects on walking stability during the workflow-relevant period are not well understood. This study quantified the phase-specific effects of three commercially available shoulder-assist exoskeletons on dynamic stability during the post-lift walking segment of a simulated lift-walk-place task. Participants performed repeated trials under four randomized conditions: no exoskeleton (NE) and three shoulder-assist exoskeleton conditions, denoted as Exoskeleton 1 (EXO1), Exoskeleton 2 (EXO2), and Exoskeleton 3 (EXO3). Whole-body motion capture was used to estimate center-of-mass (CoM) kinematics, and anterior margin of stability (MoS) was calculated from the extrapolated center of mass relative to the anterior base-of-support (BoS) boundary. Step-level minimum anterior MoS was analyzed separately during acceleration, steady walking, and deceleration phases using linear mixed-effects models. No statistically significant exoskeleton effects were observed during acceleration or steady walking. During deceleration, both EXO2 and EXO3 produced statistically significantly more negative minimum anterior MoS values than NE, whereas EXO1 did not differ significantly from NE. These findings suggest a phase-specific pattern in minimum anterior MoS, with the clearest condition-related differences observed during gait deceleration in short post-lift walking.

Keywords: Biomechanics, Occupational Safety and Health, Physical Ergonomics, Shoulder-Assist Exoskeleton

INTRODUCTION

Manual material handling (MMH) is common in occupational settings and is associated with a high risk of musculoskeletal disorders (Hartvigsen et al., 2018). To mitigate physical demands during work, exoskeletons have been increasingly introduced in industrial contexts and have shown promise for reducing localized physical workload during lifting and related tasks (Chen et al., 2023; Chen et al., 2024; Chen et al., 2025; de Looze et al., 2016; Yin et al., 2024). However, real work activities rarely consist solely of isolated lifting. Instead, workers often transition immediately from lifting to short-distance load-carrying or walking, during which balance control and dynamic stability are essential concerns relevant to safe task performance. Although occupational exoskeletons are increasingly adopted, their influence on worker stability during workflow-relevant transitions remains insufficiently understood (De Bock et al., 2022).

Dynamic walking stability is frequently quantified using the margin of stability (MoS), which integrates both center-of-mass position and velocity via the extrapolated center of mass and expresses stability relative to the base-of-support boundary (Hof et al., 2005). In gait research, MoS is typically assessed at specific gait events or summarized by the minimum value within a step to identify the most stability-challenging moment. This metric has been widely adopted as an interpretable indicator of dynamic stability across various walking paradigms (Watson et al., 2021). Previous studies have demonstrated that shoulder-support exoskeletons can influence gait performance and stability during level walking (Park et al., 2022). Notably, short overground walking bouts are inherently nonstationary due to the inclusion of gait initiation and termination, during which forward speed changes rapidly. As a result, stability measures may systematically differ across acceleration, steady walking, and deceleration phases (Orendurff et al., 2008). Despite these considerations, few studies have investigated whether commercially available shoulder-assist exoskeletons influence phase-specific walking stability during the brief post-lift load-carrying period typical of MMH tasks.

Therefore, this study aimed to quantify the phase-specific effects of three commercially available shoulder-assist exoskeletons on dynamic stability during the post-lift walking segment of a simulated lift-walk-place task. The anterior margin of stability was assessed using step-level minimum values during acceleration, steady walking, and deceleration phases, and device-related differences were compared to a no-exoskeleton (NE) condition.

METHOD

Participants

Eight adults participated in this study (age: 31 ± 3.8 years; height: 176.4 ± 7.9 cm; body mass: 78.83 ± 17.5 kg). All participants were able to walk independently without assistive devices. Participants reported no history of arm injury or chronic back pain and no known musculoskeletal, neurological, or vestibular disorders that could affect balance or gait. Additional exclusion criteria included lower-limb injury within

the past few months, prior surgery affecting normal gait, or any condition that would limit safe participation in lifting and walking tasks. All participants provided written informed consent prior to participation, and the study was reviewed and approved by the New Jersey Institute of Technology Institutional Review Board (IRB protocol #2403041907) in accordance with applicable federal regulations for the protection of human subjects (See 45 C.F.R. part 46.114; 21 C.F.R. part 56.114).

Experimental Protocol

Participants completed a simulated MMH task consisting of repeated lift-walk-place cycles while carrying a rigid box-shaped load. At the start of each cycle, the participant lifted the box from a fixed floor location, turned toward a shelving unit, and walked forward at a self-selected comfortable speed while carrying the load. After reaching the destination, the participant came to a stable stance and placed the box onto the shelf. The participant then retrieved the box, returned it to the original location, and placed it back on the floor to complete one round-trip workflow repetition. The box weighed 30 lb, and its dimensions were 14 in × 10 in × 6 in. Each cycle consisted of five complete round-trip workflow repetitions. Participants completed the protocol under four conditions: NE, EXO1, EXO2, EXO3. Two cycles were completed for each condition, resulting in 10 workflow repetitions per cycle and 40 total repetitions per participant.

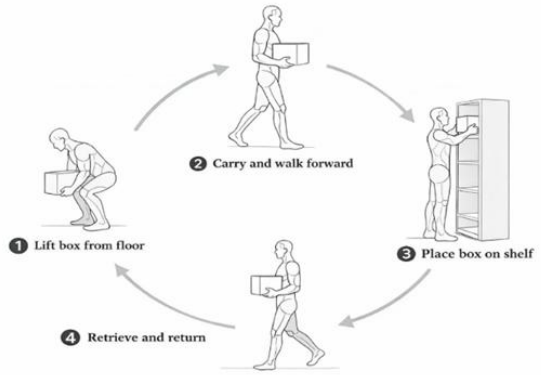


Figure 1: Schematic illustration of the experimental protocol for one cycle of the manual material handling task

Instrumentation and Kinematic Processing

Whole-body kinematics were recorded using an OptiTrack motion-capture system (NaturalPoint, Inc., Corvallis, OR, USA) at 100 Hz. Participants wore a full-body reflective marker set, and marker trajectories were labeled and processed in Motive 3 software (NaturalPoint, Inc., Corvallis, OR, USA) Whole-body center-of-mass (CoM) kinematics were estimated in OpenSim v4.4 (SimTK, Stanford University, Stanford, CA, USA) using a subject-specific scaled musculoskeletal model. Marker trajectories exported from Motive 3 were imported into OpenSim 4.4 for model scaling and inverse kinematics, and the whole-body CoM trajectory was then computed from the modeled segment kinematics and exported for post-processing in MATLAB (MathWorks, Natick, MA, USA).

Walking phase identification and MoS computation

CoM position time series were processed in MATLAB at 100 Hz. CoM velocities were calculated using a four-point central-difference scheme shown in Equation 1 and 2.

$$v(k) = \frac{-x(k+2) + 8x(k+1) - 8x(k-1) + x(k-2)}{12\Delta t} \quad (1)$$

$$a(k) = \frac{-x(k+2) + 16x(k+1) - 30x(k) + 16x(k-1) - x(k-2)}{12(\Delta t)^2} \quad (2)$$

where, $x(k)$ represents the CoM position at the k-th time sample, k is the discrete time-sample index, Δt is the sampling interval, and $v(k)$ and $a(k)$ represent the estimated CoM velocity and acceleration at the k-th sample, respectively. Horizontal walking bouts were identified from the resultant CoM speed computed from the mediolateral and anteroposterior CoM velocity components. The speed signal was low-pass filtered using a zero-phase fourth-order Butterworth filter with a cutoff frequency of 4 Hz (Buckley et al., 2020). A preliminary walking mask was defined using a speed threshold of 0.12 m/s. Short walking segments (< 0.8 s) and short stopping periods (< 0.5 s) were removed, and the resulting mask was expanded by 0.30 s to improve robustness near gait transitions. Continuous walking bouts were then identified from the processed walking mask.

Within each walking bout, locomotor phases were classified as acceleration, steady walking, and deceleration using a subject-specific speed threshold. For each bout, the threshold was defined as 70% of that participant's maximum walking speed during the corresponding walking segment. The frame at which walking speed first reached this threshold was defined as the transition from acceleration to steady walking, and the first subsequent frame at which walking speed dropped below this threshold was defined as the transition from steady walking to deceleration. Accordingly, samples before the threshold-reaching frame were labeled as acceleration, samples between the threshold-reaching and threshold-dropping frames were labeled as steady walking, and samples after the threshold-dropping frame were labeled as deceleration. If the threshold was never reached, the entire bout was classified as acceleration; if the threshold was reached but not dropped below again, the remaining bout was classified as steady walking.

Heel-strike (HS) and toe-off (TO) events were detected separately for each limb using the velocity of the midpoint between the heel and toe markers, representing the foot center. Specifically, the velocity of this foot-center point was used to identify candidate gait events, and event timing was refined using the anteroposterior velocities of the heel and toe markers within a local search window. The relevant kinematic signals were low-pass filtered at 7 Hz using a zero-phase fourth-order Butterworth filter. For each limb, stance intervals were defined from HS to the subsequent TO, and left and right single-support periods were identified as intervals during which only one foot was in contact.

Anterior margin of stability (MoS) was computed from the extrapolated center of mass (XCoM), defined as CoM position plus CoM velocity divided by the pendulum eigenfrequency, $\sqrt{g/l}$, where g is gravitational acceleration and l is leg length, shown in Equation 3 and 4. At each frame, the anterior BoS was

defined using toe-marker position. During left or right single support, the boundary was taken as the toe marker of the stance limb; otherwise, the more anterior toe marker was used according to walking direction. Frame-level anterior MoS was then calculated as the signed distance between the anterior base-of-support boundary and XCoM.

$$XCoM = P_{CoM} + \frac{v_{CoM}}{\sqrt{g/l}} \quad (3)$$

$$MoS = BoS - XCoM \quad (4)$$

To obtain a step-level stability measure, all HS events from both limbs were combined and consecutive HS events were used to define step intervals. For each step, the minimum anterior MoS value within the valid walking portion of that step was extracted to represent the most stability-challenging instant. Each step was then assigned to the acceleration, steady walking, or deceleration phase according to the phase label at the step midpoint. These phase-specific step-level minimum MoS values were used for subsequent statistical analyses.

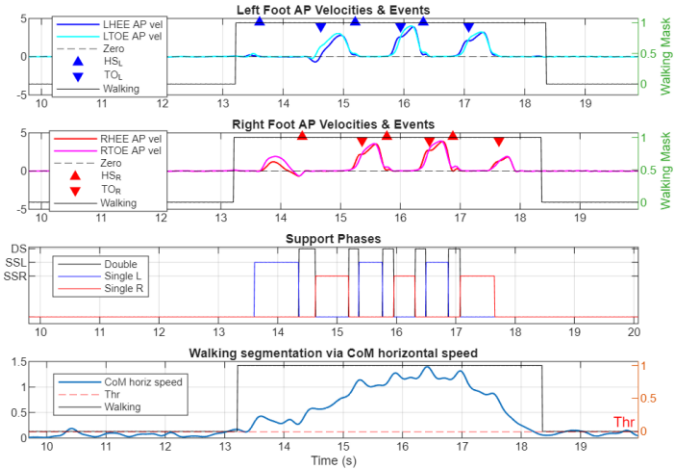


Figure 2: Example of walking-bout analysis using four panels. The first and second panels present the AP velocities of the left and right feet, respectively, together with the detected gait events (heel strike and toe off). The third panel shows the temporal classification of support phases, including single-stance and double-stance periods. The fourth panel displays the subject's horizontal CoM velocity, which served as the basis for identifying the walking bout.

Statistical Analysis

A linear mixed-effects (LMM) modeling framework (Gałecki & Burzykowski, 2013) was used to test whether minimum anterior MoS differed across exoskeleton conditions while accounting for repeated measurements within subjects through a random intercept for subject. Separate models were fitted for the acceleration, steady walking, and deceleration phases, with exoskeleton condition included as a fixed effect. For each phase, an omnibus fixed-effect test was first used to evaluate whether exoskeleton condition had a significant overall effect on minimum anterior MoS. When appropriate, post hoc pairwise comparisons between conditions were conducted based on the model-estimated marginal means. To control the family-wise error rate associated with multiple comparisons, Holm's step-down procedure was applied to adjust p-values. Statistical significance was defined as $p < 0.05$.

RESULTS

Phase-specific linear mixed-effects model results for minimum anterior margin of stability (MoS) are shown in Figure 3 and summarized in Table 1. During the acceleration phase, the estimated mean minimum anterior MoS values were similar across conditions, and no statistically significant pairwise differences were observed after Holm correction. A similar pattern was found during steady walking, and no post hoc comparison remained statistically significant after multiple-comparison correction.

In contrast, statistically significant condition-related differences emerged during deceleration. The estimated mean minimum anterior MoS values were -0.3329m for NE, -0.3995m for EXO1, -0.4095m for EXO2, and -0.4066m for EXO3. Holm-corrected post hoc comparisons showed that EXO2 and EXO3 had significantly more negative minimum anterior MoS values than NE, whereas EXO1 did not differ significantly from NE. No statistically significant differences were observed among the exoskeleton conditions themselves. Overall, these results suggest that condition-related differences in minimum anterior MoS were more apparent during deceleration than during acceleration or steady walking.

Table 1: Phase-specific summary of LMM-estimated model

Phase	Task	LMM mean	95% CI	Adjusted *p* vs NE
Accel	NE	-0.2648	0.0273	—
	EXO1	-0.2740	0.0282	1.000
	EXO2	-0.2728	0.0333	1.000
	EXO3	-0.2522	0.0321	1.000
Steady	NE	-0.4564	0.0434	—
	EXO1	-0.4235	0.0424	0.191
	EXO2	-0.4325	0.0460	0.783
	EXO3	-0.4467	0.0450	1.000
Decel	NE	-0.3329	0.0558	—
	EXO1	-0.3995	0.0675	0.080
	EXO2	-0.4095	0.0586	0.007*
	EXO3	-0.4066	0.0677	0.042*

DISCUSSION

The present study examined phase-specific changes in minimum anterior MoS during the post-lift walking segment of a simulated manual material handling task with and without the use of different shoulder-assist exoskeletons. The main finding was that condition-related differences were limited during acceleration and steady walking but became more apparent during deceleration, where EXO2 and EXO3 showed significantly more negative minimum anterior MoS values than the no-exoskeleton condition. These findings suggest that the walking termination phase may be more sensitive than earlier phases for detecting device-related changes in forward dynamic stability during short load-carrying bouts.

One possible explanation is that deceleration requires greater regulation of forward progression as the participant

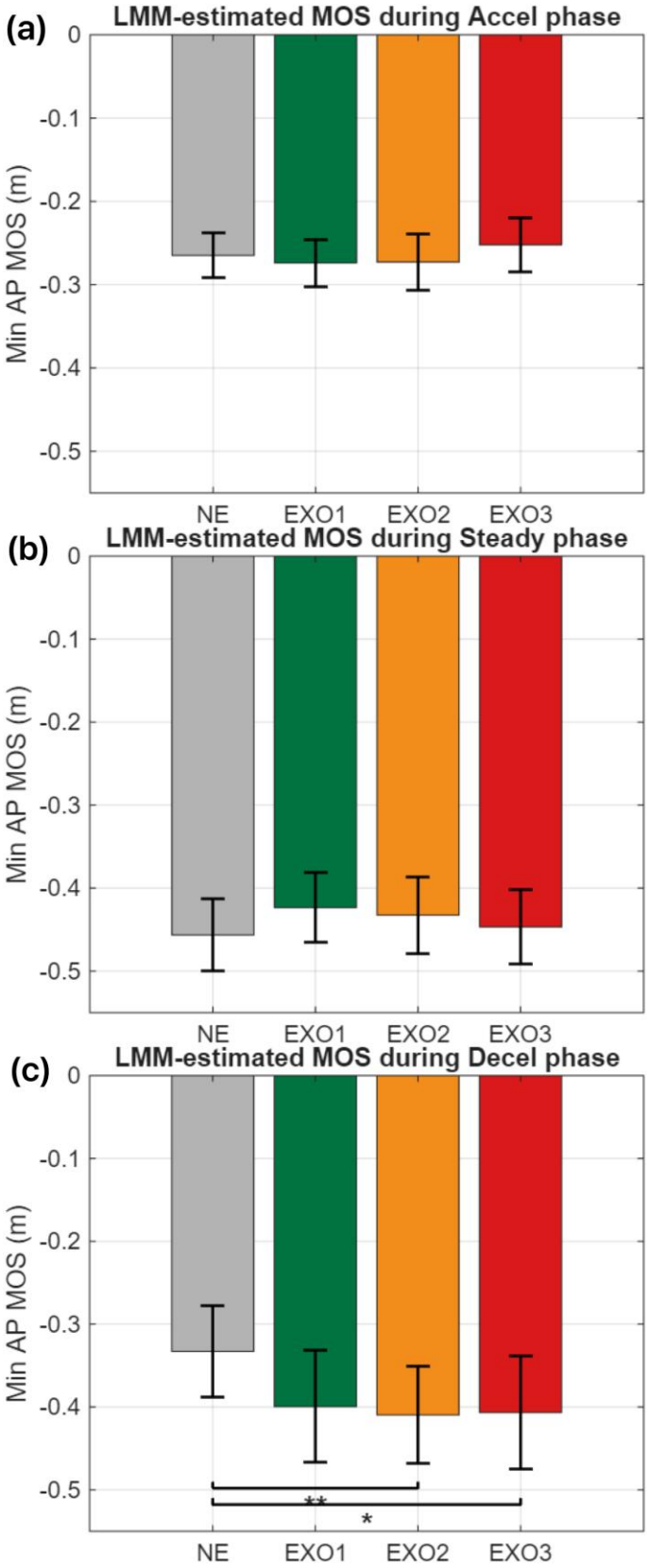


Figure 3: Linear mixed-effects model estimated mean minimum anterior margin of stability (MoS) across exoskeleton conditions during (a) acceleration, (b) steady walking, and (c) deceleration. Error bars indicate 95% confidence intervals.

approaches the destination and prepares for task completion. Unlike acceleration or mid-bout steady walking, deceleration involves reducing forward momentum while maintaining control of body position and load carriage. Under these conditions, interactions between the user, the carried load, and the exoskeleton may become more evident. Because the present task involved short overground bouts immediately following lifting, the gait pattern was inherently nonstationary, and phase-specific analysis may therefore provide more meaningful information than a single pooled stability metric across the entire walking segment.

At the same time, the present findings should be interpreted cautiously. Minimum anterior MoS reflects only one aspect of dynamic stability and should not be taken as a comprehensive measure of overall gait safety or balance performance. The present analysis focused specifically on step-level minimum anterior MoS, and other potentially relevant outcomes were not included. In addition, phase classification depended on a speed-based segmentation procedure, and the resulting estimates may be sensitive to the selected threshold and gait-event detection method. Therefore, the observed deceleration-phase differences should be viewed as a targeted finding within the current analytical framework rather than as definitive evidence of overall exoskeleton-related stability impairment.

Despite these contributions, the present study should be considered preliminary. The sample size was limited to eight participants, which reduces statistical power and limits the extent to which the findings can be generalized. Moreover, only minimum anterior margin of stability was examined, and this metric alone cannot capture all aspects of dynamic stability during load carrying. The results may also depend on the specific phase-segmentation and gait-event detection procedures adopted in this study. Accordingly, the deceleration-phase differences observed here should be interpreted cautiously. Future work with larger samples and a broader set of stability and gait measures is needed to confirm whether these phase-specific patterns persist.

CONCLUSION

This study investigated phase-specific minimum anterior margin of stability during the post-lift walking segment of a simulated manual material handling task across three commercially available shoulder-assist exoskeletons. No clear condition-related differences were found during acceleration or steady walking phase, whereas differences were observed during deceleration phase. These results indicate that phase-specific analysis may provide useful information when evaluating exoskeleton effects during short load-carrying tasks. Given the limited sample size and the focus on a single outcome measure, the present findings should be considered preliminary. Further work with larger samples and additional gait and stability measures is needed to determine whether the observed deceleration-phase pattern is robust.

Disclosure Statement

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the National Institute for Occupational Safety and Health

(NIOSH), Centers for Disease Control and Prevention (CDC). Mention of any company or product does not constitute endorsement by NIOSH/CDC.

Declaration of Conflicting Interests and Funding

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. This work was partially supported by the National Institute for Occupational Safety and Health (NIOSH) under contract #75D30123P17823.

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